

## Part 02: General

### 1. Differences

- **Batch:** A group of SQL statements sent to the database engine and executed together as a single unit.
- **Script:** A saved .sql file containing a sequential set of SQL commands and batches.
- **Transaction:** A logical unit of work that must fully succeed (COMMIT) or completely fail (ROLLBACK).
- **BackUp:** A saved copy of database files used to restore data in case of failure or loss.

### 2. Transaction Logging

- **Meaning:** Recording all data modifications into a log file (.ldf) before writing them to the main database file.
- **Why:** Ensures system recovery after crashes, maintains data consistency (ACID), and enables point-in-time restore.

### 3. Soft Delete vs. Hard Delete

- **Soft Delete:** Marks a record as deleted using a flag (e.g., IsDeleted = 1) while keeping it on disk.
- **Hard Delete:** Permanently removes the record from the database table using the DELETE command.

## Part 03: Bonus

### 1. Self-Study Report

- A research document created by a student to cover, summarize, and demonstrate understanding of an advanced topic.

### 2. Trigger

- **Definition:** A special stored procedure that automatically runs when a specific event (INSERT, UPDATE, DELETE) occurs.
- **Why Use It:** Enforces complex business rules, tracks audit history, and automates dependent database tasks.

### 3. Permissions (GRANT & REVOKE)

- **Permissions:** Access rights given to users to perform specific operations (SELECT, INSERT, UPDATE) on database objects.
- **GRANT:** Syntax to give access rights to a user (GRANT SELECT ON Employees TO Mohamed;).
- **REVOKE:** Syntax to remove previously granted access rights from a user (REVOKE SELECT ON Employees FROM Mohamed;).



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